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To cite this article: Lorena G. Barberia & Natália de Paula Moreira (27 Jun 2025): Executive approval and accountability: the impact of global shocks in less developed democracies, Journal of Elections, Public Opinion and Parties, DOI: [10.1080/17457289.2025.2507100](https://doi.org/10.1080/17457289.2025.2507100)

To link to this article: <https://doi.org/10.1080/17457289.2025.2507100>



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Executive approval and accountability: the impact of global shocks in less developed democracies

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ABSTRACT

This article offers an original theory which elucidates how voters in less developed settings evaluate macroeconomic performance and fiscal policies. We argue that voters face a critical challenge in the attribution of responsibility in an era of global market integration. We then present a theory where the complex interaction between the economy, government policies, and executive approval is understood by considering that there are substantial differences between ordinary times, and periods in which large-scale, global exogenous shocks hit the domestic economy. In ordinary times, voters evaluate incumbents for economic performance and policy. During large-scale exogenous shocks, however, voters continue to assess incumbent competence, but the task of making performance and policy-based evaluations is much harder as it depends on a more sophisticated understanding of the source of the shock, and what types of fiscal and monetary responses might be more appropriate to mitigate the effect of the crisis. We test our assumptions and theoretical expectations with monthly data for Brazil from 1997 to 2022.

ARTICLE HISTORY Received 24 October 2023; Accepted 6 January 2025

Introduction

The extent to which developing democracies effectively sanction bad performers in polls throughout an incumbent's mandate has been argued to be a powerful tool for greater accountability and development. However, scholars disagree about whether international conditions that strongly impact economic performance also largely explain incumbent approval in less developed economies (Leigh 2009). This disagreement revolves around whether the mass public in developing countries rewards (punishes) incumbents under a good (bad) economy. On one side of the spectrum, researchers claim

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 Supplemental data for this article can be accessed online at <http://dx.doi.org/10.1080/17457289.2025.2507100>.

that the relatively low levels of political understanding among the mass public in developing countries result in exogenous external shocks undermining political accountability (Campello and Zucco 2016; Leigh 2009; Zucco and Campello 2020). An alternative view suggests that democratic accountability in less developed settings can coexist with global market integration (Baker 2003; Carlin and Hellwig 2020).

Central to this debate is a dispute about whether voters respond to booms and busts that coincide with global shocks. Recessions in emerging economies are often more complex than the view presented in most textbooks to describe how these episodes function in developed economies (Carrière-Swallow and Céspedes 2013). Resource constraints matter. Fiscal space is reduced, inflation is harder to control, informal sectors are significant, and financial markets are shallower (Kose and Ohnsorge 2021). Vulnerable populations rely even more on state policies to buttress them against these shocks. Even though existing research has devoted attention to why these episodes are distinct from an economic viewpoint, there is less consensus on whether these extraordinary economic circumstances are distinct from “ordinary” times for incumbents and democratic accountability in developing democracies.

In this study, we argue that given the wild swings in domestic economic performance, the high levels of social inequalities, limited development of financial markets and the relative lack of social safety nets, there should be substantial political responsiveness to economic performance and fiscal policy in less developed settings. But, we also argue that a key to accurately modelling such relationships is to control for periods when large-scale external shocks hit the national economy during a recession. We thus advance a theory that identifies the macroeconomic contexts in which we should expect voters in less developed settings to hold executives responsible for macroeconomic performance and fiscal policy. Our argument focuses on economic accountability during these major recessions versus periods with ordinary business cycle fluctuations (Blanchard and Watson 1984). Major recessions occur when steady declines in growth change its expected long-term trend. These have coincided with global shocks in developing democracies (Rand and Tarp 2002). We show that this occurs even in large economies such as Brazil; as a result, these events produce additional stress on accountability.

Brazil is a valuable case study for our theoretical argument. A relatively large developing economy, we present evidence showing that the timing of recent recessions coincides with periods of large-scale global economic disruption. Greater globalisation (and the trade agreements and openness that often come with it) make a country more susceptible to exogenous shocks. Still, the magnitude of these shocks is comparatively smaller in larger economies (Rodrik 2011). To the extent we can show that exogenous shocks affect executive accountability, our results have implications for

developing countries more generally, as most countries are much smaller in relative terms.

There are also important institutional factors that underscore why it is crucial to test this theory for Brazil. As in many democracies in Latin America, presidents play the lead and dominant role in driving fiscal policies (Focanti, Hallerberg, and Scartascini 2016). For this reason, Brazil represents a context with significant clarity of responsibility for economic and fiscal policy. Furthermore, the country's fiscal situation is representative of that of developing countries with middle incomes where Ardanaz, Hallerberg, and Scartascini (2020) find a strong link between fiscal consolidations and electoral behaviour.

Finally, Brazil represents a critical case from the viewpoint of democratic accountability. It is home to a large electorate that is highly vulnerable to economic shocks and has experienced substantial gains in literacy in the last century (Schwartzman 2004). There is a free and independent press, and the overwhelming majority of voters seek information and follow media reporting on current events (Latin American Public Opinion Project (LAPOP) 2022). Three-fourths of the electorate turnout to vote in elections. Research shows that compulsory voting tends to increase citizens' attention to news and their efforts to discuss politics in their social networks (Carreras 2016). Since the transition from autocratic rule in the late 1980s, power has been transferred between presidents from opposing political parties.

We test our theoretical expectations with monthly data on executive approval for Brazilian presidents from 1997 to 2022. Consistent with our theory, we find that, in ordinary times, incumbents who preside over relatively better (worse) macroeconomic outcomes are rewarded (punished) in executive performance evaluations. Holding macroeconomic performance constant, presidential approval ratings are negatively affected by fiscal consolidations in these periods. In contrast, when domestic recessions coincide with large-scale global shocks, we find that declines in macroeconomic performance and contractionary fiscal policy do not impact executive approval ratings.

The rest of the paper proceeds as follows. In the next two sections, we briefly review the two bodies of literature that motivate our paper. The first group of studies focuses on debates about accountability and incumbent approval in less developed settings in the context of globalisation. The second set of studies concerns the extent to which responsibility attribution occurs in developing economies. We then introduce our argument and outline our theoretical expectations for how large-scale global shocks affect the responsiveness of voters. The following section presents the results of our hypothesis tests that show that presidential approval ratings are impacted by macroeconomic performance and fiscal policy in 'ordinary' times. In contrast, when domestic recessions coincide with large-scale

global shocks, we find that a poor-performing macro economy or tighter fiscal policies do not impact executive approval ratings.

Democratic accountability and the global economy

Students of development have long been interested in understanding how economic performance affects incumbent presidents in less developed settings, but most studies have focussed their attention on elections (Benton 2005; Cole, Healy, and Werker 2012; Gélinau 2013; Murillo and Visconti 2017; Remmer 1991, 2014; Singer 2013). These studies have generally concluded that there is democratic accountability, especially during elections when incumbents are punished by worsening economic performance and policies. Although there has been less research on how the economy influences executive approval, the research that has been conducted suggests that there is a strong and fairly robust relationship between economic performance and the popularity of executives in less developed economies (Carlin, Love, and Martinez-Gallardo 2015; Echegaray and Elordi 2001; Kelly 2003; Stokes 1996).

More recently, however, the theory that executive approval in less developed countries responds to economic performance has come under fire (Campello and Zucco 2016; Carlin and Hellwig 2020; Zucco and Campello 2020). According to one viewpoint, globalisation produces perverse effects on economic voting. Scholars have argued that accountability is absent (Campello and Zucco 2016; Zucco and Campello 2020) in low-savings-commodity-exporting democracies because the ability of voters to attribute responsibility to incumbents becomes blurred as less developed countries come to depend on international markets. The empirical evidence that exogenous shocks impact presidential performance is interpreted as confirmation that “voters do not separate chance from competence when evaluating their presidents (Campello and Zucco 2016, 600).”

The conclusions drawn by these recent studies of less developed countries contrast strikingly with research on democratic accountability and economic voting in advanced democracies. In the developed world, it has been argued that voters are pretty sophisticated in their analytical abilities and that citizens assign responsibility for economic performance to incumbent executives conditional on both the domestic political context (Lewis-Beck 1997; Powell and Whitten 1993) and international conditions (Duch and Stevenson 2010; Kayser and Peress 2012; Powell and Whitten 1993). In developed countries, voters have long been theorised to evaluate incumbent competence by assessing national deviations from national (Alesina and Rosenthal 1995) and international economic performance (Powell and Whitten 1993). Studies have argued that voters employ a competency model (Alesina and Rosenthal 1995) in which comparisons relative to past economic performance

contribute to how voters evaluate the incumbent's ability to manage the economy. Duch and Stevenson (2010), for example, argue that voters are knowledgeable about shocks to the economy and whether these shocks are of domestic or foreign origin. They adapt the competency model to propose a theory in which voters compare the variance of economic outcomes across countries to extract a signal about incumbent competence influencing their vote. Similarly, Kayser and Peress (2012) show that the media play a crucial role by providing information to voters in developed democracies, enabling them to develop cues that can be used as benchmarks to assess national incumbent performance relative to peer economies.

Moreover, voters in advanced economies have also been shown to evaluate incumbents based on the fiscal policy performance of their governments. Besley and Case (1995) develop a model showing that voters evaluate the government's fiscal policies to gauge if they are in alignment with desired macroeconomic outcomes by first assessing the state of the economy. An important collection of studies shows that voters reward and punish governments for changing the level and types of taxes, the extent and composition of taxation, and the degree and form of taxation relative to other revenue sources (e.g. borrowing and debt). These country-specific studies include Niskanen (1975, 1979), Peltzman (1992) and Geys and Vermeir (2008a) for the United States, Hibbs and Madsen (1981) for Sweden, Geys and Vermeir (2008b) for Germany, and Tillman and Park (2009) and Foucault, Seki, and Whitten (2017) for OECD countries.

Dissenting literature stresses that voters' evaluations of the economy are driven by partisanship (Evans and Andersen 2006; Marsh and Tilley 2010; Rudolph 2003). However, the empirical evidence thus far has produced evidence suggesting that the link between partisanship and economic evaluations may be weaker or in the reverse causal direction (Lewis-Beck 2006; Lewis-Beck, Nadeau, and Elias 2008). Furthermore, in the limited research directed at how economic evaluations are affected by shocks to domestic markets from the world economy, work by Hellwig, Ringsmuth, and Freeman (2008) shows that priming partisans about economic globalisation does not affect their attribution of responsibility.

Despite the recognition that voters are well-equipped to evaluate the capacity of incumbents to manage the economy in developed settings, there is a robust consensus in the literature that global shocks increase the difficulty for voters to evaluate competence (Alcañiz and Hellwig 2011; Duch and Stevenson 2010; Hellwig 2008). An example is the work of Hellwig (2008), who shows that globalisation reduces the utility of economic considerations as a signal for evaluating politicians in elections in France and Great Britain. Nevertheless, although higher exposure to global trade is recognised to have reduced the magnitude of the incumbent competency signal, Duch and Stevenson (2010) emphasise that democratic accountability

continues to operate in developed democracies integrated into global capital and trade markets.

In less developed settings, initial economic voting research emphasised that international economic performance shocks would be transmitted and their effects could be captured entirely with domestic economic performance and policy indicators (Remmer 1991, 1993). In other words, studies based their conclusions on findings drawn from analysing how domestic economic indicators affected political outcomes. Given the significant economic volatility of the international economy, the findings from this branch of research undoubtedly leave many questions unanswered. In contrast, recent research (Campello and Zucco 2016; Zucco and Campello 2020) has argued the opposite. Scholars have focussed on the relationship between exogenous global economic volatility and election or popularity indicators. Domestic policy and performance indicators have been omitted in these models, and these models also suffer from bias.

In our view, as we explain in further detail in the section on our argument, scholars must include the national and global economy to correctly test theories claiming that the world economy has obscured executive accountability in less developed settings. Furthermore, effort must be directed at accurately modelling the interaction between both. If global market forces affect domestic performance, then both factors must be employed in empirical tests (Hellwig 2008). For this reason, we model and test these complex interactions in this paper.

Responsibility attribution for economic performance and policy in less developed countries

The debate in the political economy literature on the effect of the global economy on incumbent approval depends critically on differences in how scholars view the level of sophistication of voters (Kayser and Peress 2012). In recent years, a significant point of contention has been a debate about the degree of voter understanding in so-called “new” democracies (Barberia and Avelino 2011; Brender and Drazen 2005; Shi and Svensson 2006). To date, the majority of existing research aimed at elucidating this question has tested the ability of voters to assess incumbents in the context of electoral behaviour (Ardanaz, Hallerberg, and Scartascini 2020). There has been far less attention focussed on how voters in low-income democracies respond to policy change over the length of an administration.

In recent transitions from authoritarian rule, some scholars have asserted that voters are “inexperienced” (Brender and Drazen 2005) or that there is a higher share of the “uninformed” (Shi and Svensson 2006) in developing countries. As a result, politicians engage in fiscal manipulation and secure re-election at higher rates than in advanced economies (Brender and

Drazen 2008). On the other hand, others have argued that governments in early transitional democracies have limited room to manoeuvre. Instead of engaging in fiscal expansions that appease the mass public, governments selectively target fiscal policies to win support from swing voters (Drazen and Eslava 2010). Alternatively, work by Barberia and Avelino (2011) has argued that pressures to avoid the loss of support by strategic groups of voters that might threaten regime stability help explain fiscal policy's responsiveness in developing democracies. Yet, even those who have advanced arguments about the relative naivety of voters in less developed environments readily acknowledge that voters can learn over time. For example, Brender and Drazen (2005) acknowledge that voters are naive in the early democratic transition but, subsequently, learn and adapt.

In less developed democracies, there is also recognition that increased media presence matters for accountability (Besley and Burgess 2002). In a review of experimental studies, Pande (2011) concludes that access to information changes voter evaluations of public policies, even in very low-income settings where voters lack formal education. In India (Besley and Burgess 2002) and Brazil (Ferraz and Finan 2011), studies have argued that the media provides information to voters about incumbent performance, which explains significant differences in accountability across districts.

An important strand of research has argued that voters leverage a more sophisticated analytical toolbox to assess incumbents in the context of globalisation in less developed environments. Baker (2003), for example, argues that voters are more likely to approve of trade policy liberalisation depending on economic context (periods of positive growth) and political affinity with executives (those who approve of the president are more likely to support trade liberalisation). Moreover, the author shows that voters in less developed countries support trade policy liberalisation because they weigh the benefits for mass consumers (prices, quality and variety of consumer goods) and decide that these benefits outweigh labour market considerations. In a similar vein, Kerner (2010) shows that pension reforms in many Latin American countries increased capital ownership by the mass public. As a result, he argues that an increasing share of citizens uses performance-based judgements about returns on pension capital to assess neoliberalism's track record. Alcañiz and Hellwig (2011) find that voters are sufficiently aware of their government's (lack) of political control and of their country's ties to international structures. As a result, they conclude that voters reallocate blame (and credit) contingent on the extent to which the economy is integrated into world trade and capital markets. On the other hand, Campello and Zucco (2016) argue that "citizens' access to information is generally low, and as a result of these factors, exposure to news about other economies is limited at best (591)."

Similar to advanced settings, we believe there are equally complex interactions between domestic economic performance, government policies,

and global markets in less developed settings. We share the view that has been underscored both in developed (Alcañiz and Hellwig 2011; Duch and Stevenson 2006; Hellwig 2008) and less developed environments (Campello and Zucco 2016) that voters face a critical challenge in the attribution of responsibility in an era of global market integration. However, we disagree that the lack of voter assessments in less developed countries combined with the degree of exposure to international markets results in the absence of democratic accountability in these settings. Instead, we believe there is an interaction between the domestic and global economies, and this relationship must be appropriately measured and specified in theoretical models and empirical tests. Our theoretical specification allows for an informational externality, which affects both executive accountability and incentives for incumbents to increase taxes. In the next section, we describe this theoretical argument.

Theoretical expectations and empirical strategy

Theoretically, we have reasons to believe that the attribution of responsibility in developing democracies depends on whether the economy is in ordinary times, or experiencing an extraordinary economic crisis where a domestic recession coincides with a more widespread global recession. We hypothesise that the attribution of responsibility for economic performance and fiscal policy of an incumbent government depends on whether the economy is in an extraordinary period. Following this idea, this section discusses our strategy for identifying what characterises when an economy is experiencing an exceptional global-induced recession and introduces our hypotheses.

Exogenous global shocks and domestic recessions

Brazil illustrates that global exogenous shocks coincide with major economic downturns. Even though Brazil is a relatively sizeable developing economy, periods of crisis in other countries coincide with domestic recessions. We used the OECD's Composite Leading Indicators Database (OECD 2024) to identify the recession duration in the Eurozone OECD, the United States, Brazilian, Chilean, Mexican, and 5 largest Asian economies between 1997 and 2022. The OECD identifies the economy as in recession based on the "growth cycle" approach, where business cycles and turning points are measured and identified in the deviation-from-trend series. The recessions are calculated as the periods between peaks and troughs based on changes in GDP. We considered a global exogenous shock to coincide with a domestic recession if there is a shock in the country/region of origin that is simultaneous with a recession in Brazil.

Table 1. Recessions in Brazil and Global Shocks, 1997:1–2022:12.

Period	President(s)	Duration (months)	Chile	Mexico	Asia*	US	Euro	Global Crisis
1997:10–1999:6	FHC I and FHC II	21	X		X		X	Asian and Russian Financial Crises
2001:1–2003:8	FHC II and LULA I	32	X	X	X	X	X	European and US Recessions Argentine Convertibility Crisis
2004:10–2005:10	LULA I	13			X			SARS and Tsunami Crises in Asia
2008:6–2009:4	LULA II	12	X	X	X	X	X	Global Economic Recession
2011:5–2012:2	DILMA I	10			X		X	Eurozone Recession Commodity Price Collapse
2014:2–2016:8	DILMA I & II and TEMER	31	X		X			Russian Financial Crisis Commodity Price Shock
2020:3–2020:6	BOLSONARO	4	X	X		X	X	COVID-19 Pandemic

Source: OECD (2024).

Notes: The recession period for Brazil follows OECD's methodology based on the growth cycle from the period following the peak through the trough OECD (2024). For Chile, Mexico, Asia, the United States and the Eurozone, an "X" means that six or more months of recession were detected during the recession in Brazil. Asia refers to China, India, Indonesia, Japan, and Korea. Global shock refers to the source of the shock identified in the literature.

Table 1 confirms that the seven recessions that hit the Brazilian economy between 1997 and 2022 coincided with downturns in other countries.¹ Indeed, the Brazilian economy was in a recession that coincided with a global shock in 122 months (39 percent of the 312 months examined for this study). Furthermore, even though the initial shock was exogenous, these crises provoked protracted economic crises in Brazil.

There is a fundamental difference between ordinary times and moments when a large-scale exogenous shock is transmitted from the global economy during or causing a domestic recession in a developing economy. Large-scale global exogenous shocks trigger major disruptions. These periods elevate complexity, making it increasingly more complex for voters to evaluate the competency of incumbents and their management of the economy. In these unique and far-reaching moments, voters face increasing difficulty making performance and policy-based evaluations. Voters may continue to attempt to use economic outcomes to assess incumbent competence. Still, the task of evaluating incumbents is much more demanding as voters must parse out how much of the income reduction is due to the external shock versus government management

¹See the Supplementary Materials for the magnitude of these crises on unemployment, unexpected inflation, tax revenues, and per capita GDP growth.

of the economy and what types of fiscal and monetary responses might be more appropriate to mitigate the effect in the context of globalisation. For this reason, we propose a theory that posits that voters use information about whether the economy is in an exceptional (e.g. in a recession that coincides with a large-scale, global shock) or in an ordinary moment to evaluate economic performance and policies.

In ordinary times, recessions are interpreted as signalling the lower ability of politicians to manage the economy (Nordhaus 1975; Nordhaus, Alesina, and Schultze 1989; Rogoff 1987). Voters obtain information about the performance of the country's economy based on media coverage and then evaluate economic performance and fiscal policies Besley and Burgess (2002); Shi and Svensson (2006). In this state of the world, voters assume that low output is a consequence of a politician's lower ability to produce well-being in response to an adverse shock. In economies more intensely engaged in world trade and international capital market flows, voters observe that economic booms and busts fluctuate with global market flows. However, this does not impede their ability to assess economic performance or government policies; it only increases the variance of the competency signal (Duch and Stevenson 2010).

Two hypotheses follow:

- (H1) In ordinary times, downturns in economic performance will negatively affect executive approval holding fiscal policy constant; and,
- (H2) In ordinary times, contractionary fiscal policies will negatively affect executive approval holding economic performance constant.

During a globally-induced recession, there is greater uncertainty in responsibility attribution because voters have to identify the share of the income hit that is due to the global shock separately from the share of the decline in income due to bad government management of this shock. Global shocks increase the difficulty for voters to evaluate competence (Alcañiz and Hellwig 2011; Duch and Stevenson 2010).

Two hypotheses follow:

- (H3) When the economy experiences a large-scale exogenous shock around the time of a domestic recession, economic performance will not negatively affect executive approval holding fiscal policy constant; and,
- (H4) In domestic recessions that coincide with global shocks, contractionary fiscal policy will not negatively affect executive approval holding economic performance constant.

We have presented a theoretical framework to elucidate the conditions under which global market forces affect democratic accountability in less

developed settings. It is essential to underscore that we emphasise the asymmetric effects that adverse exogenous global shocks produce during domestic recessions. Developing democracies are also exposed to positive exogenous shocks, such as the wave of commodity price booms in the prices of oil, mining products, and, to a lesser extent, agricultural goods, that benefited Latin American exporters between 2003 and 2013. These shocks have been shown to have a limited trickle-down effect on the domestic economy (Sánchez-Ancochea 2021). These booms have an even more limited impact on economies, such as Brazilian, where agriculture is only 4 percent of GDP. Authorities are more likely to intervene to moderate demand to ensure that these expansionary shocks are calibrated before generating price inflation.

Data and methods

We now test whether politicians are punished for economic mismanagement in high versus low economic policy certainty settings. A critical test of our theoretical expectations is statistical evidence confirming that there are long-term decreases in executive approval when economic performance worsens and taxes increase in ordinary times. We expect that statistical evidence will not confirm the same pattern does not occur when large-scale exogenous shocks coincide with Brazil's recessions. We use aggregated time series data on approval for Brazilian presidents from 1997 to 2022 to test these hypotheses. In the next section, we introduce the model specification and data sources. We then discuss how the results contribute to our theoretical propositions about how voters judge presidents in exceptional and ordinary times.

Model specification and data

The dependent variable is the monthly change in presidential approval from 1997:1 to 2022:12. Our sample is of 312 monthly observations consisting of the last two years of the first term and the entire second term of Fernando Henrique Cardoso (FHC) from the *Partido da Social Democracia Brasileira* (PSDB) (from 1997:1 to 2002:12), the first (from 2003:1 to 2006:12) and second (from 2007:1 to 2010:12) terms of Luiz Inácio “Lula” da Silva, the first (from 2011:1 to 2014:12) and second terms (from 2015:1 to 2016:8) of Dilma Rousseff, both from the *Partido dos Trabalhadores* (PT), the interim government of Michel Temer from the *Partido do Movimento Democrático Brasileiro* (PMDB), and the term of Jair Bolsonaro (from 2019:1 to 2022:12).²

²Jair Bolsonaro ran as a partisan of the *Partido Social Liberal*, but he left the party following his election in November 2019. For the remainder of his term, he had no party affiliation.

The monthly data for presidential approval are from the Executive Approval Database (EAD) (Carlin et al. 2023)(see Figure 1). The EAD 3.00, released in 2023, uses Stimson's dyads-ratio algorithm (Stimson 2018) to construct a continuous latent executive approval time series with polling data from multiple academic and commercial pollsters in Brazil. The dyads-ratio algorithm estimates and helps correct house and sampling frame effects.³

To test our theory, we employ a single measure for economic performance and fiscal policy. To test how the macro economy affects incumbent approval, we use the monthly rate of the labour force that is unemployed using a unified monthly rate with data compiled by Hecksher (2020); Lameiras and Hecksher (2024).⁴ With respect to why unemployment is used to measure economic performance. In their review of economic voting, Lewis-Beck and Paldam (2000) emphasise that unemployment has moved to the centre stage in the economic voting literature as high inflation has become less severe and frequent of a problem. In the period we study, which is after the Real Plan stabilised inflation, there is evidence that inflation was a less salient issue for Brazilian voters. Furthermore, it should be emphasised that research has shown that people's knowledge about unemployment and inflation is relatively informed. Nannestad and Paldam (1997), for example, show that people know unemployment rather well as compared to inflation.

Fiscal policy is measured as federal tax revenues as a share of GDP. The taxation data were collected from the Central Bank of Brazil (as reported by the Ministry of Finance). There are good reasons why revenue reflects overall changes in tax policies in Brazil⁵. For the majority of tax instruments, the federal government does not distribute tax revenue to other levels of government.⁶ Research on taxation policies underscores that a substantial share of the variation in tax revenue is due to policy in Brazil (Orair et al. 2012). Unfortunately, the data for tax revenue are only available from

³The EAD presidential approval data highly correlates with two executive approval polling aggregator sources, which we consulted for robustness checks: El-Dash (2024) and Jota (2024). The correlation between EAD and Jota is 0.967 (p -value 0.000). The correlation between EAD and El-Dash is 0.962 (p -value 0.000).

⁴The methodology for measuring unemployment was revised by the national statistics agency (IBGE) in 2012. From 1997 to 2012, unemployment was collected on a monthly basis in the Pesquisa Mensal de Emprego (Monthly Report of Employment). Starting in 2012, the IBGE began collecting quarterly employment data in the PNAD-Continua (the Pesquisa Nacional de Amostragem de Domicílios- Continua, or National Report of Household Samples-Continuous data). Lameiras and Hecksher (2024) calculated monthly equivalent measures from 2013 to 2022.

⁵We consider fiscal expenditures to be less precise for testing our theory than revenue in the case of Brazil (Arretche 2012). Whereas the federal government is the central actor in tax revenue collection, municipal governments are important actors in executing fiscal expenditures. For example, concerning social spending, municipal governments are the key actors in health and education policies thus making the evaluation of national actors more complex.

⁶It should be noted that tax revenues are affected by factors outside the government's control. However, there is no reliable index of tax policies or rates that capture changes across multiple tax types (e.g. personal and corporate income taxes, VAT, export, etc.) on a monthly basis for Brazil. Furthermore, the tax reform data collected by Barberia and Castro (2019) confirm that different taxes were being raised and loosened in the same month.



Figure 1. Job Approval of the Brazilian President: 1997:1 to 2022:12 (%).

Source: Carlin et al. (2023).

January 1997 onward. Thus, we are unable to test our argument for earlier periods in the 1980s and 1990s.

Our preferred specification is an Error Correction Model (ECM) since a battery of unit roots suggests that approval, unemployment, and tax revenue are non-stationary.⁷ The dependent variable (Approval) is the change in the approval rating of the incumbent president (the measure depicted in Figure 1).

A core element of our theory is that the effect of economic performance and government policy on presidential approval is conditional on the presence of recessions that coincide or are exacerbated by exogenous global shocks. An implication of our theory is that when a global shock coincides with a domestic recession, a switch in the “state of the world” occurs, which produces an economic context that is distinct from ordinary times. To capture this cutpoint, we use a dichotomous measure that identifies that there are different “accountability” mechanisms in these two states of the world. In practical terms, this means that there is an interaction between a dichotomous measure for global recessions and the explanatory

⁷As severe threats to inference arise in analysing non-stationary data, we conducted a battery of unit root tests on the dependent and explanatory variables. The Augmented Dickey–Fuller tests permit the rejection that the series is nonstationary, but the Phillips-Perron, ERS, DF-GLS, and KPSS tests indicated that the data-generating process of the variables are $I(1)$. These unit root tests are reported in the Supplementary Materials for all variables in level and differenced form.

fiscal and economic performance measures.⁸ We use this specification to examine whether the short and long-term effects of unemployment and tax policy changes on presidential approval are conditional on whether the economy is in a period which coincides with a global exogenous shock or experiencing “ordinary” economic times.

Our model⁹ can be summarised in the Engle-Granger two-step form as:

$$\Delta Approval_t = \alpha_0 + \alpha_1 R_t + \alpha_2 \Delta Unemp_t + \alpha_3 \Delta Tax_t + \alpha_4 \Delta Unemp_t R_t \\ + \alpha_5 \Delta Tax_t R_t + \alpha_6 ECM + \alpha_7 PresidencyTerm_t + \alpha_8 Events_t + \epsilon_t,$$

where $ECM = (Approval_{t-1} - \beta_1 Unemp_{t-1} - \beta_2 Unemp_{t-1} R_t - \beta_3 Tax_{t-1} - \beta_4 Tax_{t-1} R_t)$, and $\alpha_6 = -(1 - \theta_1 - \theta_2 R_t)$, and $R_t = GlobalShockRecession$.

It should be noted that $\alpha_6 = -(1 - \theta_1 - \theta_2 R_t)$ allows for different error correction rates. In ordinary times, the error correction rate is $\alpha_6 = -(1 - \theta_1)$. In globally-induced recessions, the error rate becomes $\alpha_6 = -(1 - \theta_1 - \theta_2 R_t)$.

Given problems of multicollinearity, we present the most parsimonious model possible which allows us to test our argument. As explanatory variables, we include one lag of the dependent variable ($Approval_{t-1}$) and additional differenced lags of the dependent variable to eliminate serial correlation. In addition, we include dummy variables to control for each presidential term. As is the norm in presidential approval temporal dynamic studies, we include dummies for significant events representing important structural breaks. The model includes a dummy variable for the June 2013 protest when massive street mobilizations across more than 100 cities occurred throughout Brazil. President Dilma Rousseff’s job approval rating plummeted by nearly 16 percentage points. We capture this by a dummy variable equal to 1 in the period (6:2013-7:2013) and 0 otherwise. We also include a measure for the period of goodwill that presidents enjoy in their first month in office (January of the first year of the presidential term). Although dummy variables for a series of other events including a dummy variable to control for corruption scandals (e.g. *mensalão* and *LavaJato* scandals) were initially included, these measures were excluded as the estimated coefficients were not statistically different from zero.

Results

In this section, we report the results of statistical tests of our theory using an ECM model that allows us to test how incumbent approval is affected by

⁸Our specification adopts a time-dependent error decomposition. See Clarke, Ho, and Stewart (2000); Philips, Rutherford, and Whitten (2015); Pickup and Hobolt (2015) who use time-related interactive terms in an Error Correction framework. We follow these authors in their implementation of interaction estimations in ECMs.

⁹We use this notation to make the model more easily comprehensible. In the Supplementary Materials, the entire derivation is provided. It should be noted that $\alpha_6 = \gamma$ in the Supplementary Materials.

economic performance and fiscal policies, considering the presence and absence of exogenous global shocks during domestic recessions. We have argued that voters and politicians use information about whether the economy is in a globally induced crisis to evaluate economic outcomes. Since these hypotheses are directional hypotheses, we use a one-sided *t*-test of the null, with the long-term effect being less than or equal to zero, to evaluate our four hypotheses.

The complete regression results for all parameters are reported in the Supplementary Materials.¹⁰ We are most keenly interested in the long-term effects of our explanatory variables as these serve to help us understand the lasting effect of these changes on presidential approval (De Boef 2001). The coefficients on each of the lagged economic variables and the rate of adjustment (the coefficient of the lagged dependent variable) are used to calculate long-term effects that are presented in Table 2.

The ECM model suggests a low error correction rate for presidential approval in Brazil. This indicator, which is the denominator in the long-term effect calculations, is similar in magnitude to findings reported by Clarke et al. (2005) for U.S. presidents. In ordinary times, deviations in economic performance and fiscal policies are eroded at a rate of approximately 14.4% per month starting in the month after the shock occurred (or between 8.5% and 20.4% using a 95% confidence interval). In turn, the error correction rate during a domestic recession that coincides with a global disruption is lower. In these periods, deviations in economic and fiscal performance decline by approximately 17.8% per month starting in the month after the shock occurred (or between 9.3% and 26.3% using a 95% confidence interval). These results shed light on our argument; the speed of adjustment is faster in moments of crisis.

In the absence of globally induced recessions, we argued that voters would punish incumbents for downturns in economic performance and increases in taxation (H1 and H2). The results confirm that when the economy is *not* in a globally-induced recession, increases in unemployment and higher taxes are strong predictors of a president's job performance. All else equal, a one percent increase in the unemployment rate produces a 1.829% long-term decrease in presidential approval. The effect of economic performance is statistically significant at $p < 0.05$ (one-tailed test) or better. We had predicted that tax increases would weaken the president's support in ordinary times. All else equal, a one percent increase in tax revenue (as a share of GDP) during an "ordinary" month reduces presidential approval by

¹⁰The estimated coefficient on each of the first-differenced variables is used to calculate the short-term effect. We do not consider that the statistical significance test results of these parameters are sufficient to verify our theory because we understand that short-term effects are brief interruptions, after which presidential approval returns to its equilibrium relative to the other variables in the co-integrating relationship.

Table 2. Estimated Long-Term Effects of Economic Performance and Fiscal Policy on Presidential Approval in Globally-Induced Recessions and Ordinary Times.

	Ordinary Times	Global Exogenous Recession
Unemployment	–1.829** (0.845) $p = 0.016$	–1.033 (0.848) $p = 0.112$
Tax/GDP	–1.117* (0.719) $p = 0.061$	0.272 (0.727) $p = 0.646$

Note: Full ECM results are available in the Supplementary Materials. Standard errors in parenthesis. *** indicates $p < 0.01$, ** indicates $p < 0.05$, * indicates $p < 0.10$ for one-tailed tests.

1.117 percentage points in the long-term with the effect being statistically significant at $p < 0.10$ (one-tailed test).

Our results also confirm our theoretical expectations that the impact of a change in taxation on presidential approval is conditional on the absence of a globally induced recession (H3 and H4). Public opinion on the president's job performance is not sensitive to higher unemployment or higher taxation during recessions. Neither of these measures has a long-term statistically significant effect on approval.

Figure 2 shows how changes in unemployment and taxation dynamically impact presidential approval during ordinary times. The left panel of Figure 2 shows the response to presidential approval after a one-unit increase in unemployment. The right panel depicts the effect of a one-unit increase in tax revenue in $t = 0$.

During ordinary times, a one-unit increase in unemployment and tax revenue (as a share of GDP) triggers a negative response in approval. The period-specific effects occur mostly in the month after a shock ($t + 1$) and beyond. By the 12th month after the impulse, the cumulative loss of approval when there is an increase in unemployment is approximately 1.49%. During globally-induced recessions, the period-specific effects of an increase in tax revenue are mostly immediate and minor in terms of overall magnitude. The period-specific effects are negative and decay over the next months after the impulse. By the 12th month after the pulse, approval decreases by 0.959%. These results confirm that increases in unemployment and taxation trigger a statistically and substantively larger negative response in presidential support in ordinary times, and the majority of the effect occurs in the first months following the policy change.

It should be stressed that each time we add additional variables, these must occur in both states of the world and this contributes to increasing multicollinearity. We initially considered including per capita growth and inflation in the ECM on presidential approval. Both growth and inflation are considered stationary in unit root tests. Since they are not considered part of the long-term equilibrating relationship, given their stationarity, we did not include these variables in the preferred specification discussed here.

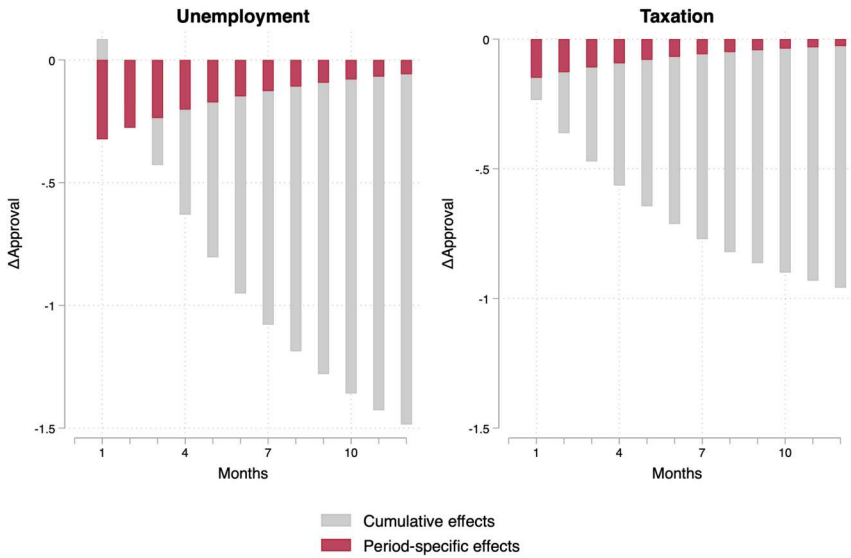


Figure 2. Estimated Lag Distributions for Presidential Approval in Ordinary Times.

Note: The figure shows the impulse response function (IRF) for presidential approval after a shock of one unit at $t = 0$ in each one of the explanatory variables. In the Supplementary Materials, we detail how we calculated these quantities.

Conclusion

In this article, we have sought to contribute to the scholarly debate on whether voters evaluate incumbent competence and whether their ability to do so is challenged by an economy's exposure to the volatility in the global economy. Earlier work in advanced democracies with more stable economic and fiscal performance has argued that voters distinguish variations due to policy ("competency shocks") from those due to exogenous shocks to the macro economy (Duch and Stevenson 2006, 2010). However, in developing democracies, it has been argued that when exogenous shocks in the economy originate from global fluctuations, the resulting domestic shocks are attributed to the competence of politicians by voters Campello and Zucco (2016); Zucco and Campello (2020). The results of our study suggest that voters in developing democracies are more similar to their counterparts in advanced economies.

There are important reasons why competence evaluations occur. Voters leverage information which permits them to distinguish between two distinct "states of the world." In the more uncertain environment, when a global shock coincides with a domestic recession, voters are reluctant to judge poor fiscal performance and tax hikes. However, in ordinary times, which we define as the absence of a large-scale global exogenous shock and which can include a domestic recession, citizens can punish incumbents for

worsening economic conditions and increases in taxes. In other words, voters in developing democracies also seek to leverage information to attribute responsibility and competence matters in ordinary times.

There are important policy implications stemming from our findings for democracies in less developed settings. During economic downturns, provided that there are identifiable global economic shocks, incumbent presidents will act strategically to raise taxes and may enact policies that contribute to higher unemployment. There is some anecdotal evidence of this in developing democracies. During the COVID-19 pandemic, for example, Gupta and Tovar Jalles (2022) find that pandemics increase the likelihood of tax reforms happening – in the 45 developing countries, whereas the reverse occurs in advanced economies.

Our findings highlight questions that warrant further investigation. For purposes of brevity, we highlight two issues. First, in the period for which monthly fiscal data are available for Brazil (e.g. 1997 to the present) and the using the definition we adopted to define a recession that coincided with a global exogenous shock (e.g. recessions in other global markets coinciding for at least six months with a recession in Brazil, all recessions were linked with global shocks. Ideally, our theory should also be tested in democracies that also experienced domestic recessions. There are democracies, for example, which remain less integrated with global markets. In these settings, it would be valuable to verify if there are differences in the share of domestic recessions that coincide with global recessions. If our theory's assumptions are correct, we should find even stronger patterns of economic voting in these countries.

Second, we have argued that there are fundamental differences in how voters assess incumbents during ordinary and negative global shocks. To appropriately test these differences, we employed an ECM where an interaction helps to identify the “two states of the world.” The testing of theories with asymmetries, such as the one advanced in this study, requires a relatively large number of time points, which constrains the ability to test these theories in countries that experienced shorter democratic spurts, or have more limited time series for economic and fiscal performance. Furthermore, there are only a limited number of studies that have estimated dynamic time series models with interactions. There are important methodological questions that need to be more fully explored on how these models should be estimated, and interpreted.

Acknowledgments

The authors thank Guy D. Whitten, Neale El-Dash, Mark Pickup, Andrew Q. Philips, Fernando Limongi, Stephen J. Kaplan, Katsunori Seki, Daniela Schettini, Gilmar Masiero, Jorge I. Domínguez, Ian Prates, Rodrigo Orair, Sergio Gobetti, Mark Hallerberg, the anonymous reviewers of the paper and participants in the “Taxation, Revenue, and Fiscal Capacity” conference at Texas A&M University and the Electoral Studies

Group and the Inequality Studies Group in the Department of Political Science at the University of São Paulo for their comments on earlier drafts of this paper. The authors thank Maria Leticia Claro, Rodrigo Sartori Lima, Ingrid Castro and Camila Gonçalves for their research assistance on this project.

Disclosure statement

No potential conflict of interest was reported by the author(s).

Funding

Natália de Paula Moreira acknowledges fellowship support from Fundação de Amparo à Pesquisa do Estado de São Paulo FAPESP (2016/13199-8, 2017/11559-0).

Data availability

Data and replication materials are available on the authors' page on the Harvard Dataverse.

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